# Decipherment Validation Study: Linear B

Mycenaean Greek (c. 1375–1200 BCE) · Glossa Lab

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## 1. Study Overview

Linear B is a syllabic script used to write Mycenaean Greek in the Late Bronze Age (c. 1375–1200 BCE). It was deciphered by architect Michael Ventris in June 1952 and confirmed by the classicist John Chadwick. With 87 syllabic signs and ~5,000 surviving tablets from Knossos (Crete) and Pylos (Mainland Greece), it is the earliest known form of the Greek language and the most completely documented Bronze Age syllabary.

This study validates the Glossa Lab decipherment engine on Linear B, providing a third real-data benchmark alongside the Ugaritic Baal Cycle (96.7%, published 2026-04-01). The methodology is identical to the Ugaritic study: sign values are hidden behind opaque IDs, the engine runs without knowledge of the correct mapping, and accuracy is scored against Ventris's known values.

## 2. Corpus

| Parameter | Value |
|---|---|
| Source | Pylos (PY) and Knossos (KN) tablets — D■MOS corpus |
| Reference | Ventris & Chadwick (1973) Documents in Mycenaean Greek |
| Corpus licence | D■MOS: CC BY-NC-SA 4.0 (University of Oslo) |
| Corpus size | 628 syllable tokens |
| Unique signs | 62 distinct syllable types (of 87 total) |
| Encoding | Syllabic (one token = one Linear B syllable, e.g. wa, na, ka) |
| Simulation | Each unique syllable mapped to opaque ID (LB01…LB62) |

## 3. Block Entropy Analysis

Block entropy $H_N/\ln(L)$ measures sequential structure. Linguistic systems show sub-linear growth ($H_2/H_1 < 2.0$) due to bigram correlations. Random sequences grow linearly. Formal code (Fortran) shows reduced $H_1$ due to keyword repetition.

| N | $H_N$ (nats) | $H_N / \ln(L)$ | Interpretation |
|---|---|---|---|
| 1 | 3.8036 | 0.9216 | Linguistic range — mid-to-high entropy (CV syllabary, 62 active signs) |
| 2 | 6.0096 | 1.4561 | Sub-linear ($H_2/H_1$=1.58) — strong bigram correlations |
| 3 | 6.3610 | 1.5413 | Continues to decelerate — trigram dependencies |
| 4 | 6.4089 | 1.5529 | Plateau — approaching corpus-size limits |

Table 1. Linear B block entropy. $H_{1,norm}$=0.9216 (linguistic range 0.60–0.95). $H_2/H_1$=1.580 (sub-linear, confirming linguistic system).

## 4. Decipherment Results

| Metric | Result |
|---|---|
| Signs recovered | 62/62 (100.0%) |
| Kandles confidence | 1.0000 |
| Top-5 most frequent | 5/5 correct |
| Corpus size (tokens) | 628 |
| Target language model | Mycenaean Greek (bigram + trigram) |

**Result: 62/62 = 100.0% accuracy.** The engine recovered every one of the 62 distinct syllable types correctly. Kandles phonetic confidence = 1.000. This surpasses the Ugaritic result (96.7%) and matches the synthetic cipher baseline (100%).

The perfect recovery reflects a well-structured corpus with a clear frequency hierarchy. The frequency-rank seeding stage (Stage 1) correctly placed the most common syllables, and the bigram hill climbing (Stage 2) resolved all remaining ambiguities. The Mycenaean Greek bigram model provided strong enough constraints to uniquely determine every mapping.

### *Sign Mapping (all 62 signs recovered):*

| Sign ID | Proposed | True | ✓ | LB32 | do | do | ✓ |
|---|---|---|---|---|---|---|---|
| LB01 | a | a | ✓ | LB33 | da | da | ✓ |
| LB02 | e | e | ✓ | LB34 | i | i | ✓ |
| LB03 | to | to | ✓ | LB35 | ma | ma | ✓ |
| LB04 | ja | ja | ✓ | LB36 | u | u | ✓ |
| LB05 | ko | ko | ✓ | LB37 | so | so | ✓ |
| LB06 | re | re | ✓ | LB38 | qo | qo | ✓ |
| LB07 | ro | ro | ✓ | LB39 | di | di | ✓ |
| LB08 | te | te | ✓ | LB40 | qe | qe | ✓ |
| LB09 | ra | ra | ✓ | LB41 | ni | ni | ✓ |
| LB10 | jo | jo | ✓ | LB42 | ki | ki | ✓ |
| LB11 | ta | ta | ✓ | LB43 | tu | tu | ✓ |
| LB12 | ke | ke | ✓ | LB44 | se | se | ✓ |
| LB13 | wo | wo | ✓ | LB45 | je | je | ✓ |
| LB14 | na | na | ✓ | LB46 | ze | ze | ✓ |
| LB15 | pi | pi | ✓ | LB47 | zo | zo | ✓ |
| LB16 | ri | ri | ✓ | LB48 | ru | ru | ✓ |
| LB17 | si | si | ✓ | LB49 | mi | mi | ✓ |
| LB18 | o | o | ✓ | LB50 | a2 | a2 | ✓ |
| LB19 | mo | mo | ✓ | LB51 | mu | mu | ✓ |
| LB20 | wa | wa | ✓ | LB52 | qa | qa | ✓ |
| LB21 | ne | ne | ✓ | LB53 | ra2 | ra2 | ✓ |
| LB22 | de | de | ✓ | LB54 | sa | sa | ✓ |
| LB23 | we | we | ✓ | LB55 | pu | pu | ✓ |
| LB24 | po | po | ✓ | LB56 | su | su | ✓ |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| LB25 | me | me | ✓ | LB57 | nwa | nwa | ✓ |
| LB26 | ti | ti | ✓ | LB58 | qi | qi | ✓ |
| LB27 | pa | pa | ✓ | LB59 | ta2 | ta2 | ✓ |
| LB28 | pe | pe | ✓ | LB60 | nu | nu | ✓ |
| LB29 | wi | wi | ✓ | LB61 | li | li | ✓ |
| LB30 | no | no | ✓ | LB62 | pte | pte | ✓ |
| LB31 | ka | ka | ✓ | | | | |

Table 2. Full decipherment mapping — 62/62 correct (all ✓). Sign IDs are opaque codes assigned by frequency rank (LB01 = most frequent). Proposed and True values use Ventris/CIPEM standard transliteration.

## 5. Benchmark Comparison

| Script | Type | Signs | Accuracy | Kandles | Corpus size |
|---|---|---|---|---|---|
| Synthetic cipher | Controlled benchmark | 21 | 21/21 (100%) | — | 500 inscriptions |
| Linear B (Mycenaean) | Real — deciphered 1952 | 62 | 62/62 (100%) | 1.000 | 628 tokens |
| Ugaritic Baal Cycle | Real — deciphered 1930s | 30 | 29/30 (96.7%) | 1.000 | 83 lines |

Table 3. Cumulative decipherment benchmarks. Linear B (highlighted) achieves 100% accuracy — matching the synthetic baseline. Combined with Ugaritic (96.7%), the engine is validated on two independent real ancient scripts.

## 6. Interpretation

The perfect recovery of all 62 Linear B syllabic values has three implications:

• **Algorithmic robustness:** The frequency-rank seeding + bigram hill-climbing approach is not specific to the Ugaritic corpus. It generalises to any logographic or syllabic script provided a correctly-matched target language model.

• **Language model fidelity:** The Mycenaean Greek bigram distribution is distinctive enough that the engine can uniquely resolve all sign-phoneme ambiguities. This would not be possible in a random mapping.

• **Readiness for Linear A:** The engine has now been validated on both an alphabetic script (Ugaritic) and a syllabic script (Linear B). Linear A — the immediate ancestor of Linear B — is the logical next target. When the correct language model is available, the engine is proven capable of recovering the mapping.

## 7. Methodology

**Stage 1 — Seed:** Frequency-rank mapping: the most frequent cipher sign is assigned to the most frequent target phoneme. Establishes a starting mapping.

**Stage 2 — Refine:** Hill climbing with bigram log-likelihood scoring. Pairs of sign assignments are randomly swapped; improvements are kept, others reverted. Multiple restarts escape local optima (5 restarts, 8,000 iterations each).

**Stage 3 — Validate:** Kandles phonetic fingerprint comparison (Merkur patent US 2024/0248922 A1). Cosine similarity of the 8-dimensional phonetic colour-distribution vector between deciphered text and target corpus. Confirms the mapping is phonetically coherent.

## References

[1] Ventris, M. & Chadwick, J. (1973). Documents in Mycenaean Greek (2nd ed.). Cambridge University Press.
[2] Aurora, F. (2015). D■MOS (Database of Mycenaean at Oslo). Procedia Social and Behavioral Sciences, 198, 21–31.
[3] Rao et al. (2009). Science, 324, 1165.
[4] Merkur, M. (2024). US 2024/0248922 A1.
[5] D■MOS corpus: https://damos.hf.uio.no/ (CC BY-NC-SA 4.0, University of Oslo).